We talked by phone, as our schedules couldn't be meshed for an in-person meeting. She described herself as a bit of a physics geek, and said that her character, Brand, is expected to know the physics cold—and then she launched into a series of surprisingly technical physics questions: What is the relationship of time to gravity? Why do we think there might be higher dimensions? What is the current status of research on quantum gravity? Are there any experimental tests of quantum gravity? . . . Only at the end did she let us wander off subject, to music, in fact. She played trumpet in high school; I played sax and clarinet.

During the filming of *Interstellar*, I was on set very, very little. I was not needed. But one morning Emma Thomas toured me through the *Endurance* set—a full-scale mockup of the *Endurance* spacecraft's command and navigation pod, in Stage 30 at Sony Studios.

It was tremendously impressive: 44 feet long, 26 feet wide, 16 feet high, suspended in midair; able to shift from horizontal to nearly vertical; exquisite in detail. It blew me away, and piqued my curiosity.

"Emma, why build these enormous, complex sets, when the same thing could be done with computer graphics?" "It's not clear which would be cheaper," she responded. "And computer graphics can't yet produce the compelling visual details of a real set." Wherever possible, she and Chris use real sets and real practical effects, except for things that can't actually be shot that way, like the black hole Gargantua.

On another occasion, I wrote dozens of equations and diagrams on Professor Brand's blackboards, and watched as Chris filmed in the Professor's office with Michael Caine as the Professor and Jessica Chastain as Murph.[1] I was astonished by the warm and friendly deference that Caine and Chastain showed me. Despite having no role in the filming, I was notorious as *Interstellar*'s real scientist, the guy who inspired everyone's best effort to get the science right for this blockbuster movie.